MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548330 (CONVERTER, RANGE RATE D/A 4A17)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 and applicable drawings comprise the complete requirements for the range rate D/A converter used in Computer, Range CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-50652	Circuit Card Assemblies, Radar Set AN/VPS-2 General Specification for

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

F10548330	Range Rate D/A Converter

PACKAGING DATA SHEET

10548330	Packaging of Circuit Card Assembly, Range Rate D/A Converter

FSC: 1285

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings, and packaging data sheets, required by contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing FI0548330 and drawings pertaining thereto.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. As specified in MIL-C-50652.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. For the requirements of 3.5.5 through 3.5.8 logic input levels shall be as follows:

Logic 0: 0.0 +0.5 volts direct current (Vdc) level.
Logic 1: +2.8 to 5.4 Vdc level.

3.5.1 Input voltages. The input voltages shall be as specified in Table I. All voltages shall be regulated with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Bias voltage. The bias voltage measured at TP4 of the range rate D/A converter assembly shall be within the limits of +2.0 and +2.2 Vdc.

3.5.3 -310M/sec reference. The reference voltage measured at P1-12 and P1-14 shall be adjustable by R4 between the limits of -9.90 and -10.10 Vdc.

3.5.4 +310M/sec reference. The reference voltage measured at P1-16 shall be within the limits of +9.90 and +10.10 Vdc.

TABLE I. Input voltages

Voltage	Terminals
+5.0 ±0.3 Vdc	P1-30, P1-31, P1-32
+5.0 V return	P1-24, P1-25, P1-26
+15.0 ±0.3 Vdc	P1-36, P1-37
±15.0 V return	P1-22, P1-23, P1-34, P1-35
-15.0 ±0.3 Vdc	P1-20, P1-21

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3.5.5 Positive gain adjust. With the logic levels applied as specified in step 1 of Table II, the output voltage at P1-41 shall be adjustable by R15 between the limits as specified in step 1 of Table II.

3.5.6 Negative gain adjust. With the logic levels applied as specified in step 2 of Table II, the output voltage at P1-41 shall be adjustable by R16 between the limits as specified in step 2 of Table II.

3.5.7 Range rate voltages. With the logic levels applied sequentially for steps 3 through 12 as specified in Table II the range rate output voltages at P1-41 shall be within the limits as specified in Table II.

3.5.8 Range rate ratios. With the logic levels applied sequentially as specified in Table III, the ratio of the range rate output voltage at P1-41 to the reference output voltage terminal specified in Table III, shall be within the limits as specified in Table III.

3.6 Workmanship. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface blisters, excess flash, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing the base material. The assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

3.7 First article inspection. This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

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TABLE II. Range rate voltages

Step	Equivalent 1/ Range rate	Input logic levels (Reference P1 terminals)								Output voltage (Reference terminal P1-41)	
	Meters/Sec(M/S)	52	2	6	4	44	46	42	51	Not less than	Not more than
1	+310	1	1	1	1	1	1	1	0	+9.900	+10.100
2	-310	0	0	0	0	0	0	1	1	-9.900	-10.100
3	+160	0	0	0	0	0	0	1	0	+4.990	+ 5.090
4	+ 80	0	0	0	0	0	1	0	0	+2.470	+ 2.570
5	+ 40	0	0	0	0	1	0	0	0	+1.210	+ 1.310
6	+ 20	0	0	0	1	0	0	0	0	+0.580	+ 0.680
7	+ 10	0	0	1	0	0	0	0	0	+0.265	+ 0.365
8	+ 5	0	1	0	0	0	0	0	0	+0.107	+ 0.207
9	+ 2.5	1	0	0	0	0	0	0	0	+0.028	+ 0.128
10	0	0	0	0	0	0	0	0	0	-0.050	+ 0.050
11	-160	0	0	0	0	0	0	1	1	-4.910	- 5.010
12	0	1	1	1	1	1	1	1	1	-0.050	+ 0.050

1/ Range rates are listed by their schematic value, a rounded value, except for the +310 and -310 which is the true maximum range rate for which this board is adjusted.

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TABLE III. Range rate ratios

	Equivalent range rate	Input logic levels (Reference P1 terminals)								Range rate ratio		Ref term	Ratio mode
Step	Meters/Sec (M/S)	52	2	6	4	44	46	42	51	Not less than	Not more than	P1	
1	+310	1	1	1	1	1	1	1	0	+0.9802	+1.0202	16	+
2	-310	0	0	0	0	0	0	0	1	-0.9802	-1.0202	14	-
3	+160	0	0	0	0	0	0	1	0	+0.4940	+0.5141	16	+
4	+80	0	0	0	0	0	1	0	0	+0.2446	+0.2596	16	+
5	+40	0	0	0	0	1	0	0	0	+0.1198	+0.1323	16	+
6	+20	0	0	0	1	0	0	0	0	+0.0574	+0.0687	16	+
7	+10	0	0	1	0	0	0	0	0	+0.0263	+0.0369	16	+
8	+5	0	1	0	0	0	0	0	0	+0.0106	+0.0209	16	+
9	+2.5	1	0	0	0	0	0	0	0	+0.0028	+0.0129	16	+
10	0	0	0	0	0	0	0	0	0	-0.0050	+0.0050	16	+
11	-160	0	0	0	0	0	0	1	1	-0.5141	-0.4940	14	-
12	0	1	1	1	1	1	1	1	1	-0.0050	+0.0050	14	-

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL-C-50652.

4.3 Inspection provisions. As specified in MIL-C-50652.

4.4 Special sampling. As specified in MIL-C-50652.

4.5 Inspection equipment. As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. As specified in MIL-C-50652.

4.5.2.3 Test equipment. In conjunction with 4.5 the following test equipment listed in Table IV shall be utilized in the performance of the applicable tests as specified in 4.6.3.

4.6 Test methods and procedures.

4.6.1 Environmental. As specified in MIL-C-50652.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. The range rate D/A converter assembly shall be tested as follows: Connect range rate D/A converter and test equipment as shown in Figure 2. Switches S1 through S8, on the test fixture, item 7, are used to supply the logic level input signals as specified in Tables II and III. These switches supply a logic 1 in position 1 and a logic 0 in position 2.

4.6.3.1 Input voltages. Using power supplies items 1 through 3, apply input voltages as specified in Table I.

4.6.3.2 Bias voltage. Using the volt-ratio meter, item 4, in the voltage mode, measure the bias voltage TP4 of the range rate D/A converter to be as specified in 3.5.2. Use TP5 on the test fixture, item 5, as ground reference.

4.6.3.3 -310 M/sec reference. Using the volt-ratio meter, item 4, in the voltage mode, measure the -310 volt reference at TP1 and TP2 on the test fixture to be as specified in 3.5.3. Use TP5 on the test fixture, item 5, as ground reference.

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TABLE IV. Test equipment list

Item number	Description
1. Power supply	5.0 ±0.3 Vdc, regulated; ac ripple 50 mV peak-to-peak maximum; capacity, 200 milliamps (mA).
2. Power supply	15.0 ±0.3Vdc, regulated; ac ripple 50 mV peak-to-peak maximum; capacity, 200 mA
3. Power supply	Same as item 2.
4. Digital volt-ratio meter	Visual display: 5 digits Dc range: 0.0000 to ±999.99 V. Accuracy ±10.01% of reading ±0.0001% of full scale. Dc input resistance: Greater than 10 megohms. Dc ratio: 1.0999 to 1, ±0.005% of reading ±0.002% of full scale with ±10 V minimum reference. Common mode rejection: Dc greater than 134 dB, 60 Hz greater than 120 dB. Response time: 1500 milliseconds (ms) maximum.
5. Test fixture	As specified in Figure 1. See Table V for test fixture parts list.

TABLE V. Test fixture parts list

Designation	Nomenclature and description
E1 thru E4	Binding post: H.H. Smith 257 or equivalent.
J1	Connector: any of the following or equivalent. (1) 5893232, Malco Mfg. Co., Inc., Chicago, Illinois (2) 500946-01, National Connector Corp., Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
S1 thru S8	Switch SPDT
TP1 thru TP5	Tip jack: H.H. Smith 1501 or equivalent.

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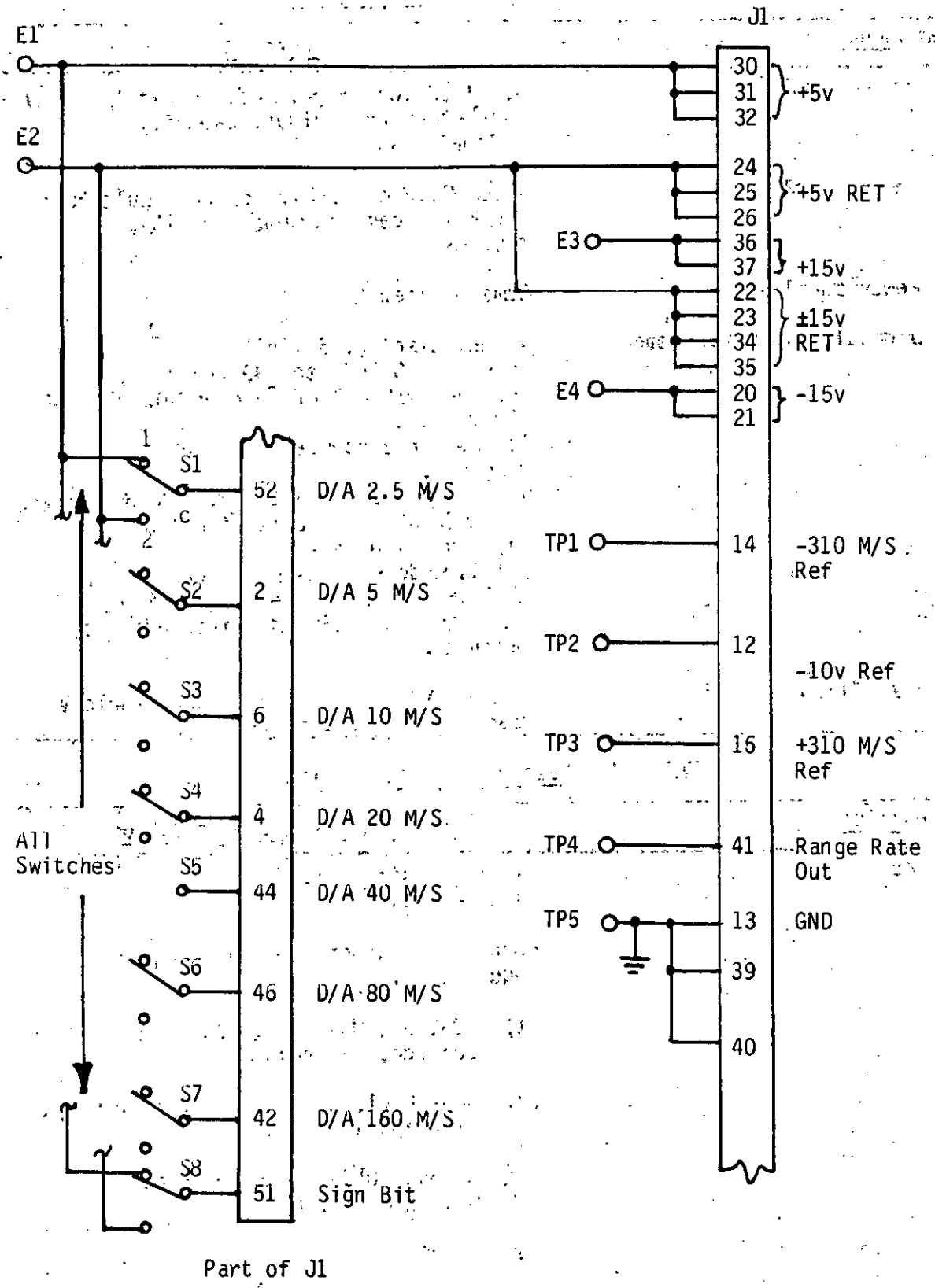


FIGURE 1. Test fixture setup.

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4.6.3.4. +310 M/sec reference. Using the volt-ratio meter, item 4, in the voltage mode, measure the +310 volt reference at TP3 on the test fixture to be as specified in 3.5.4. Use TP5 on the test fixture, item 5, as ground reference.

4.6.3.5 Positive gain adjust. Using switches S1 through S8 on the test fixture, item 5, apply the logic levels as specified in step 1 of Table II. Using the volt-ratio meter, item 4, in the voltage mode, measure the positive gain adjust at TP2 on the test fixture to be as specified in 3.5.5. Use TP5 on the test fixture as ground reference.

4.6.3.6 Negative gain adjust. Using switches S1 through S8 on the test fixture, item 5, apply the logic levels as specified in step 2 of Table II. Using the volt-ratio meter, item 4, in the voltage mode, measure the negative gain adjust at TP2 on the test fixture to be as specified in 3.5.6. Use TP5 on the test fixture as ground reference.

4.6.3.7 Range rate voltages. Using switches S1 through S8 on the test fixture, item 5, sequentially apply the logic levels as specified in steps 3 through 12 of Table II. Using the volt-ratio meter, item 4, in the voltage mode, measure the range voltages at TP2 on the test fixture to be as specified in 3.5.7. Use TP5 on the test fixture as ground reference.

4.6.3.8 Range rate ratios. Using switches S1 through S8 on the test fixture, item 5, sequentially apply the logic levels as specified in the steps of Table III. Using the volt-ratio meter, item 4, in the ratio mode, measure the range rate ratios to be as specified in 3.5.8. Connect the volt-ratio meter leads to the test fixture test points as follows:

+ voltage lead - TP4
- voltage lead - TP5
+ reference lead - TP1 for P1-14 (+ ratio mode)
TP3 for P1-16 (- ratio mode)
- reference lead - TP5

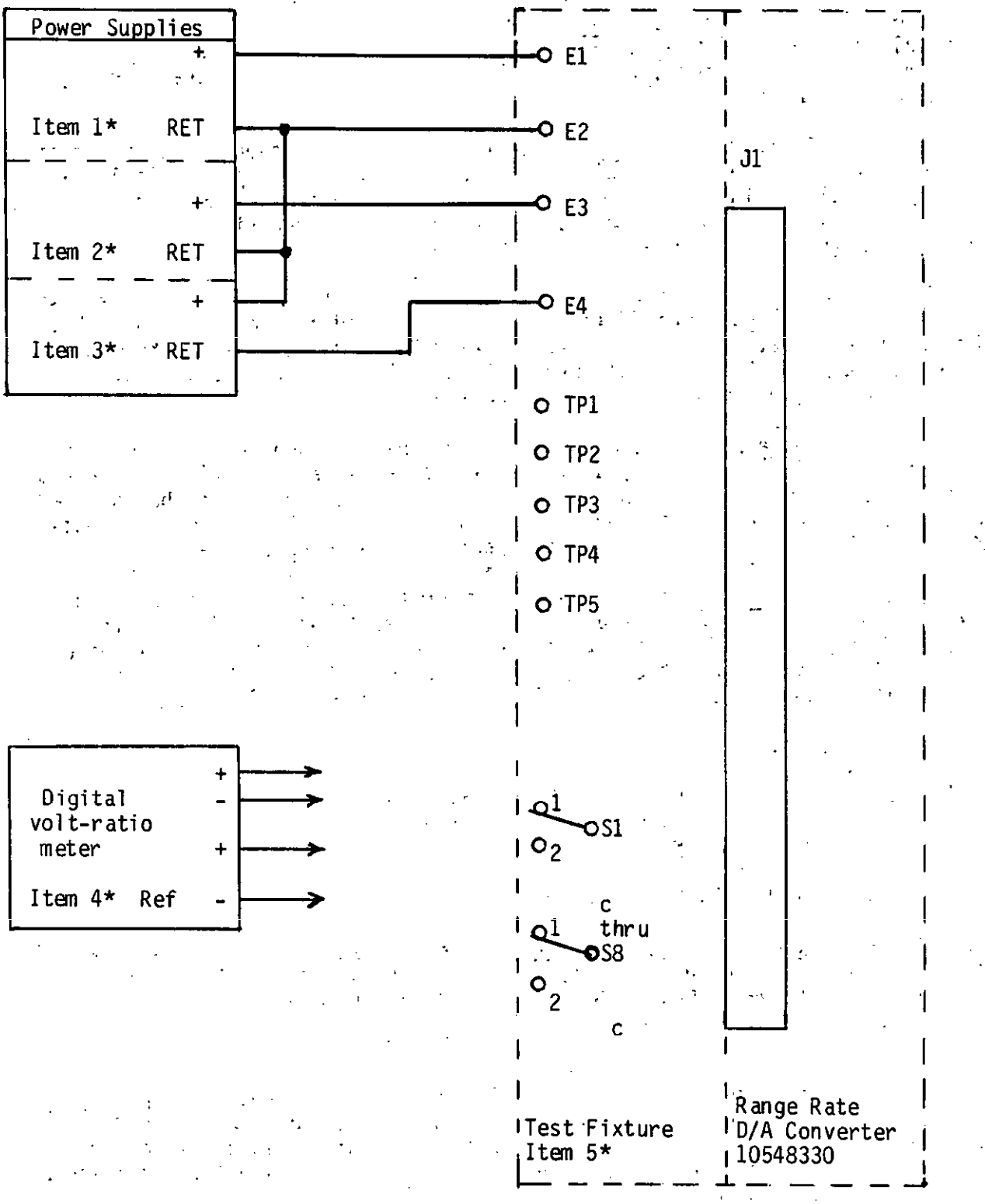
5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10548330. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The range rate D/A converter is used in Computer, Range CP-888/VPS-2, part of Radar Set AN/VPS-2. The range D/A converter receives the range rate word in binary form and converts it to an analog voltage representing range rate. The converter also furnishes two reference voltages of -10 Vdc and +10 Vdc.

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*Refer to 4.5.2.3

FIGURE 2. Range rate D/A converter test setup.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

6.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF-I, Dover, NJ 07801.

When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.).
c. The designs remaining to be submitted and the expected date of transmittal.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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